PROJECT – SWIFT SHIP TRACKER

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SWIFT SHIP TRACKER

User Story:

User Story

Swift Ship Tracker is designed to provide a simple and organized way to manage parcel deliveries. The system allows customers to submit shipment details and follow the progress of their parcels throughout the delivery process. Delivery staff can access the required shipment information and update the progress based on the current delivery stage. The system keeps the shipment information organized so that support staff can easily respond to customer enquiries. Administrators can monitor the overall delivery activities and keep track of ongoing shipments. By bringing these activities together, the system helps make parcel management easier, clearer, and more efficient.

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1. INTRODUCTION

1.1 Project Overview

Swift Ship Tracker is a digital parcel management system developed to make shipment handling more organized and convenient. The system provides a structured way to manage parcel information from the initial booking stage until the delivery is completed. It helps users maintain shipment details, follow delivery progress, and access the required information when needed. The application also supports delivery staff in managing their assigned shipments and updating their progress. By maintaining the important shipment activities in one system, Swift Ship Tracker helps improve the overall parcel management process.

1.2 PURPOSE:

The purpose of Swift Ship Tracker is to:

- Provide a structured platform for managing parcel delivery activities.
- Simplify the process of recording and handling shipment information.

- Help customers view the progress of their parcels during delivery.
- Enable delivery staff to maintain and update assigned shipment details.
- Improve coordination between customers, delivery teams, and support staff.
- Reduce dependency on manual methods for maintaining parcel records.
- Help administrators monitor shipment activities in an organized manner.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Customer / Sender	send a parcel conveniently	shipment details are difficult to manage	shipment details are difficult to manage	confused

PS-2	Customer / Sender	know my parcel progress	delivery updates are not always clear	shipment movement can change during transportation	concerned

PS-3	Delivery Agent	manage assigned parcels	handling multiple deliveries is challenging	delivery information changes frequently	pressured

PS-4	Customer	receive delivery information	updates may not reach me at the right time	communication can be delayed	uncertain
PS-5	Administrator	monitor shipment activities	tracking multiple deliveries can be difficult	operational information needs to be organized	concerned

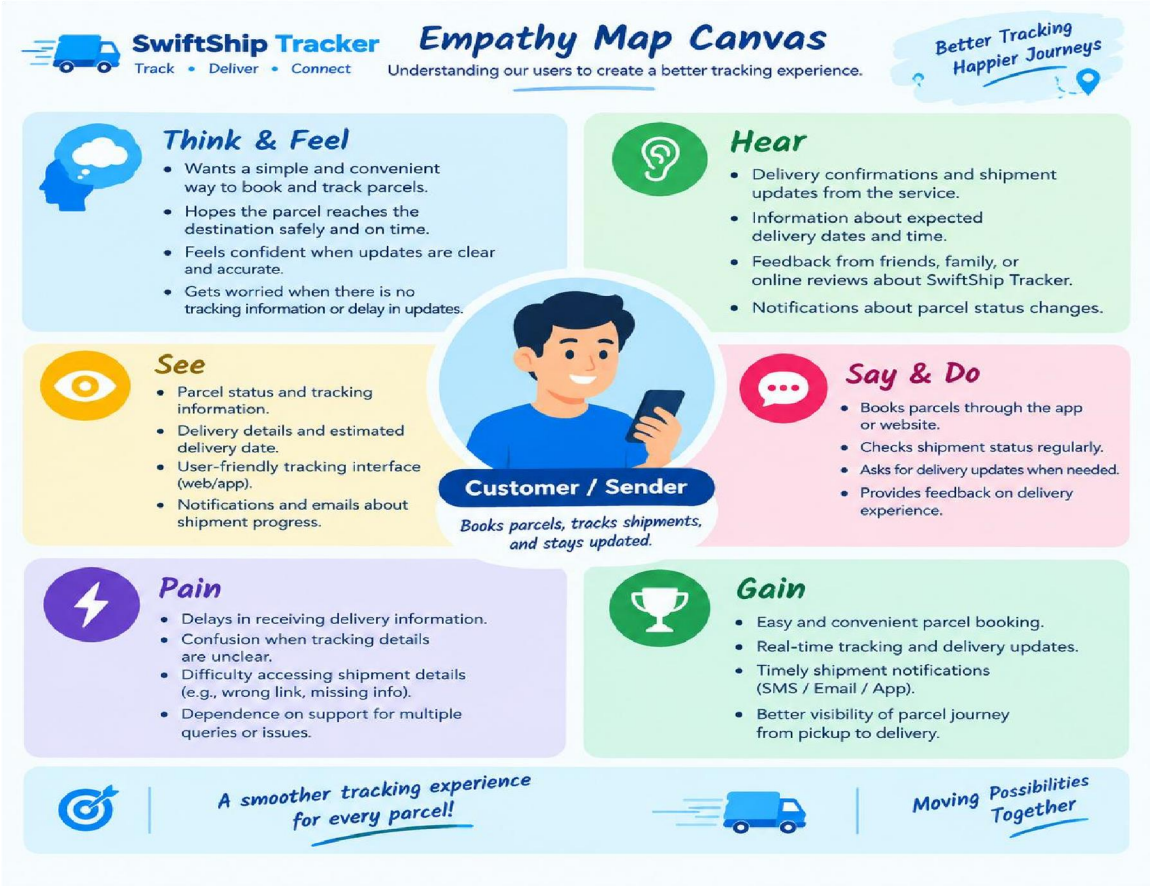


2.2 Empathy Map Canvas

2.2 Empathy Map Canvas

User Persona: **Customer / Sender**

- Think & Feel: Wants a quick and reliable way to book parcels and follow their delivery progress.
- Hear: Receives shipment updates and delivery-related messages from the service.
- See: Parcel status, shipment details, delivery progress, and expected delivery information.
- Say & Do: Books shipments, checks parcel status, and follows delivery updates. ● Pain: Unclear tracking information, delayed updates, and difficulty managing shipment details.
- Gain: Easy parcel booking, clear tracking information, quick status updates, and timely delivery notifications.



2.3 Brainstorming & Idea Prioritization

Ideas Considered

- Manual parcel registers
- Spreadsheet-based shipment tracking
- Basic online tracking system
- Salesforce CRM with AgentForce AI

Final Selection

Salesforce CRM selected due to:

- Centralized shipment information
- Automated parcel status updates
- Easy real-time tracking
- AI-based customer assistance

• Secure user access
• Reports and dashboards

3. PHASE 1 : Requirement Analysis

3.1 Business Requirements

The Swift Ship Tracker System is designed to support the day-to-day activities involved in parcel booking, shipment tracking, and delivery management. The system provides a centralized Salesforce environment for maintaining parcel, sender, receiver, delivery, and tracking information.

The major business requirements of the system are:

- Maintain parcel information including Parcel ID, parcel status, weight, and estimated delivery date.
- Maintain complete sender information such as name, email, phone, and address..
- Maintain receiver information including name, email, phone, and delivery address.
- Maintain delivery information including current location, delivery status, and estimated delivery date.
- Connect parcels with the appropriate sender and receiver records.
- Associate delivery details with the corresponding parcel.
- Allow delivery agents to update parcel status during the delivery process..
- Provide customers with parcel tracking and delivery status information.
- Automate parcel status updates using Salesforce Flows..
- Provide AI-powered parcel tracking assistance using AgentForce..
- Send automated notifications and updates during important delivery stages..
- Maintain accurate and up-to-date parcel and delivery information.
- Provide reports for monitoring parcel status and delivery performance..

3.2 Functional Requirements

The functional requirements define the major functions that the Swift Ship Tracker should perform.

FR No	Requirement
FR-1	Create and manage parcel booking records.
FR-2	Generate and maintain unique Parcel ID
FR-3	Store sender and receiver details
FR-4	Record parcel weight and delivery information.
FR-5	Maintain parcel delivery status
FR-6	Track current delivery location
FR-7	Allow delivery agents to update shipment details
FR-8	Automate parcel processing using Salesforce Flow
FR-9	Provide delivery notifications and status updates

FR-10	Provide delivery notifications and status updates
FR-11	Generate operational reports and monitoring dashboards
FR-12	Provide controlled access to parcel information for different users.

3.3 Non-Functional Requirements

NFR	**Description**
NFR-01	**Usability:** Simple and user-friendly interface for parcel management and tracking
NFR-02	**Security:** Controlled access to shipment and customer information
NFR-03	**Availability:** Accessible through the Salesforce cloud environment

NFR-04	**Reliability:** Consistent storage and retrieval of parcel and delivery data
NFR-05	**Maintainability:** Flows, configurations, and AI components can be modified when business requirements change.
NFR-06	**Scalability:** Supports increasing numbers of parcels, customers, and deliveries
NFR-07	**Performance:** Quick response during parcel search, status updates, and tracking.
NFR-08	**Data Integrity:** Maintains accurate and consistent relationships between parcel, sender, receiver, and delivery records

3.4 Data Model

The Swift Ship Tracker System uses a connected Salesforce data model to organize the information required for parcel booking, delivery, and shipment tracking**.**

The system contains the following custom objects:

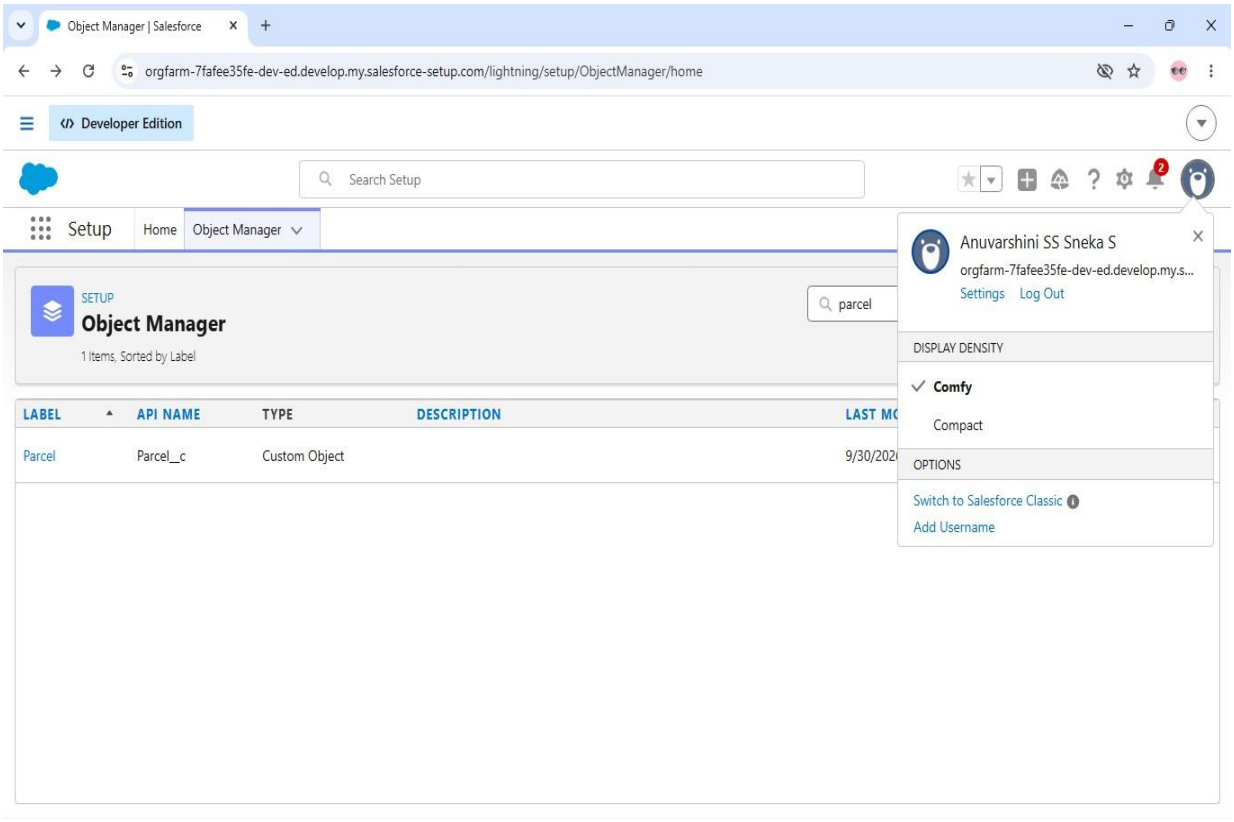
- **Parcel** – Stores core parcel details such as Parcel ID, Status, Weight, and Estimated Delivery Date.
- Client – Stores client contact and location information.
- **Delivery** – Stores delivery information such as Current Location, Estimated Delivery Date, and linked Parcel details..
- **Sender** – Stores sender information such as Address, Contact Number, and Email.
- Receiver – Stores receiver information such as Address, Contact Number, and Email.

Object Relationships

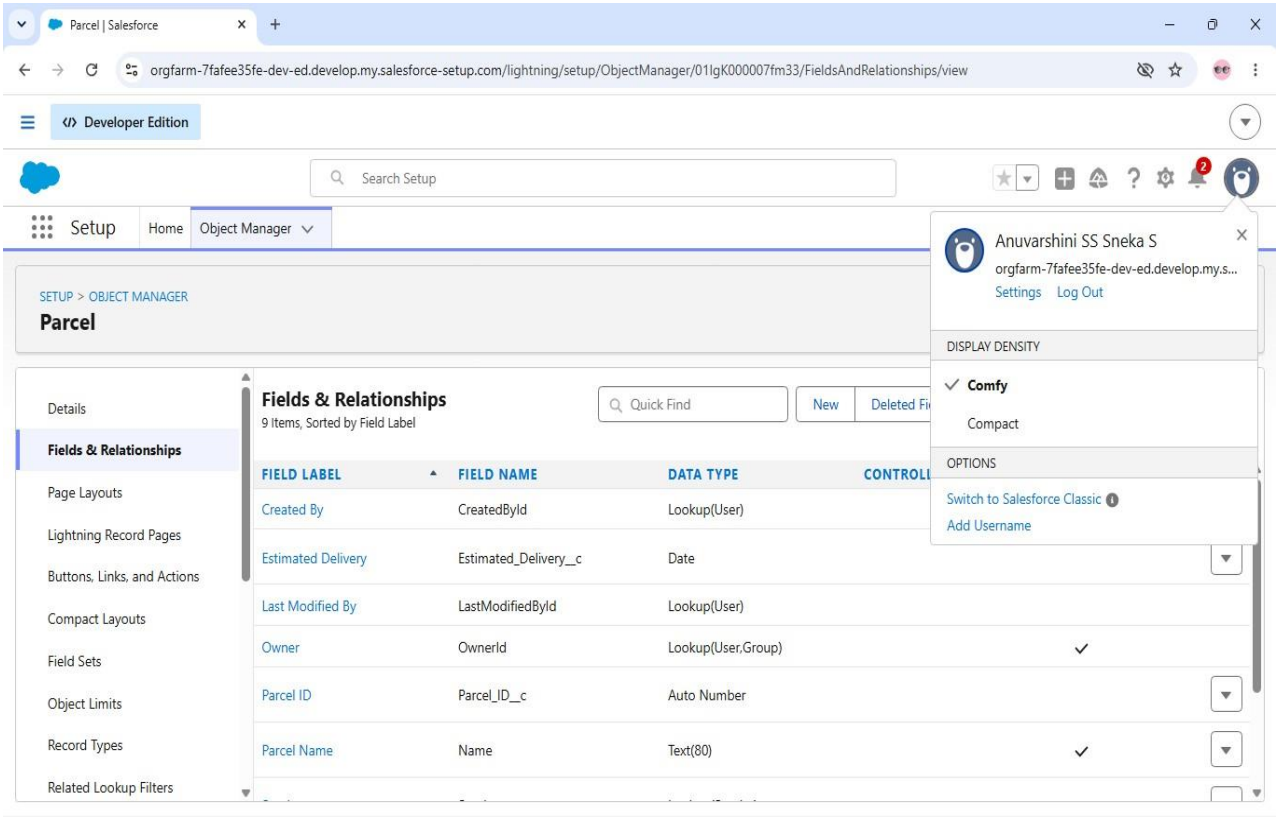
- Parcel → Sender: Lookup relationship used to associate a parcel with the sender.
- **Parcel → Receiver:** Lookup relationship used to associate a parcel with the receiver.
- **Delivery → Parcel:** Lookup relationship used to connect delivery information with the corresponding parcel.
- Delivery → Sender: Lookup relationship used to connect delivery information with the sender.
- **Receiver → Sender:** Lookup relationship used to associate receiver information with the sender.
- **Receiver → Parcel:** Lookup relationship used to associate receiver information with the corresponding parcel

4. PHASE-2 -: Salesforce Development –Backend & Configurations

Custom Objects

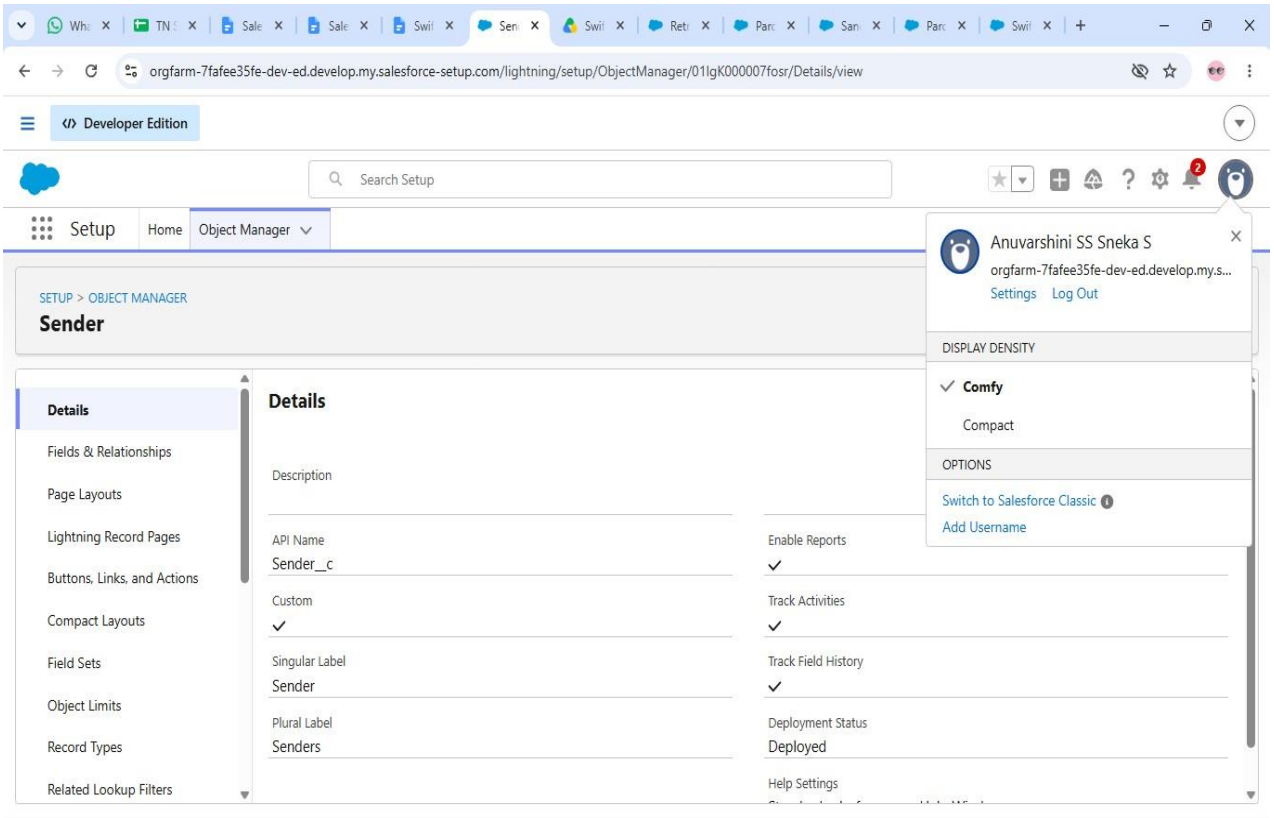


CUSTOM OBJECT



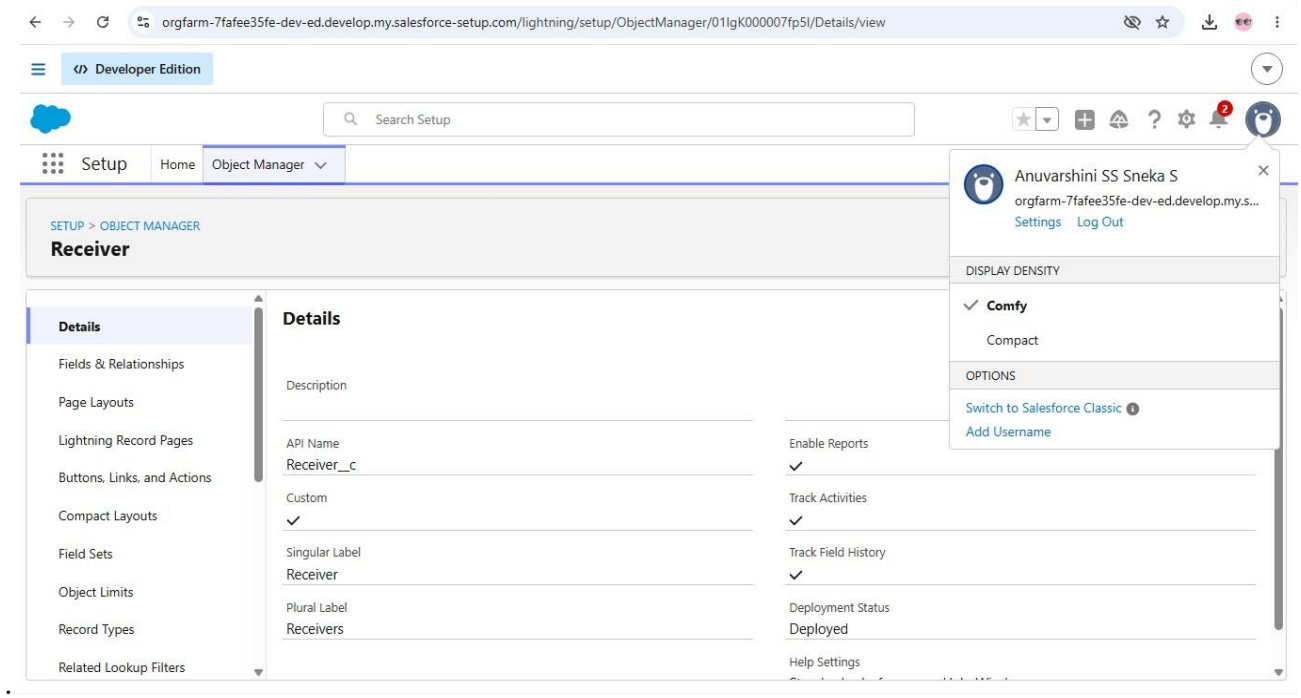
Sender

The **Sender** object is used to maintain sender information related to parcels. It stores details such as **Sender Name, Sender Email, Sender Contact, and Sender Address**. The Sender object is linked with the **Parcel** and **Delivery** objects so that each shipment can be associated with the corresponding sender.



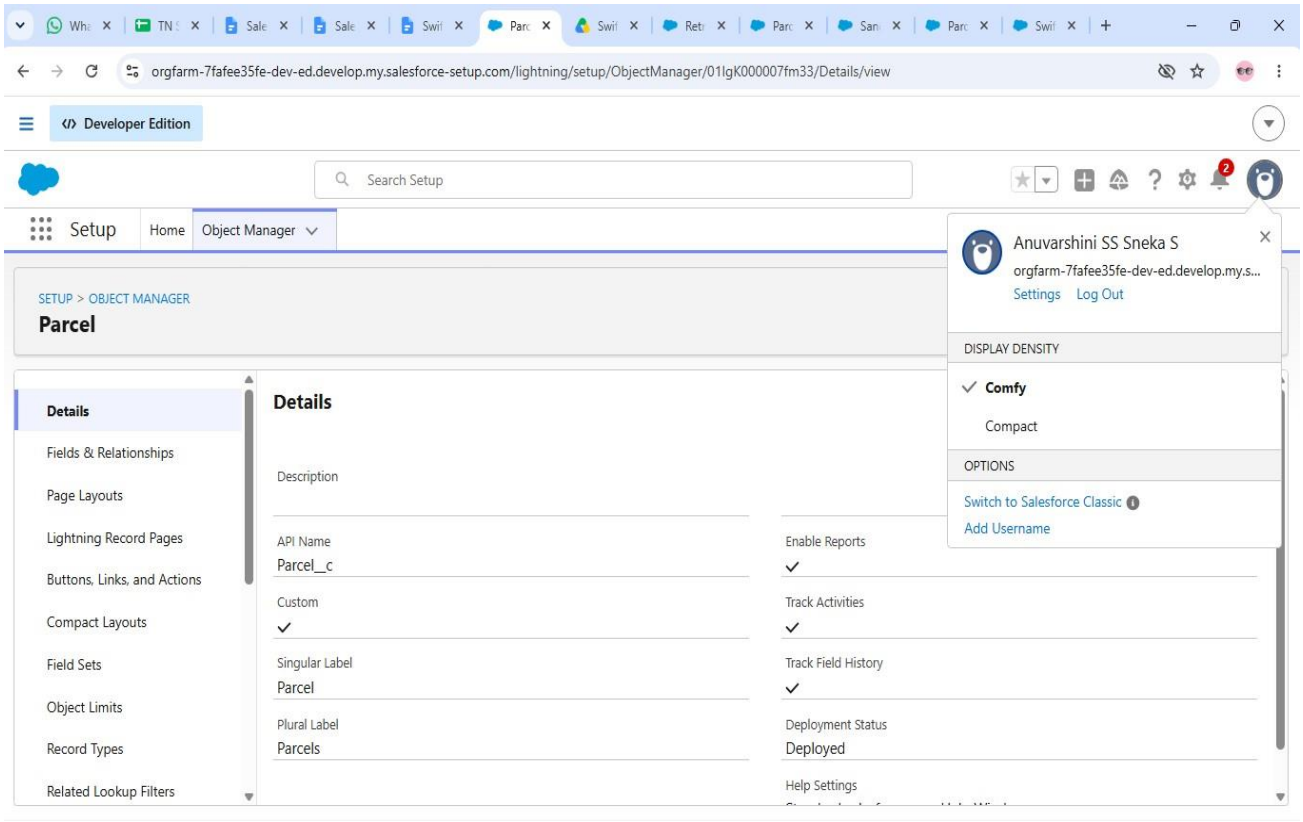
Receiver

The **Receiver** object is used to maintain information about receivers who receive parcels. It contains **Receiver Name, Receiver Email, Receiver Contact, and Receiver Address**. Receiver records are associated with the **Parcel** and **Sender** objects so that each shipment can be connected with the corresponding receiver.



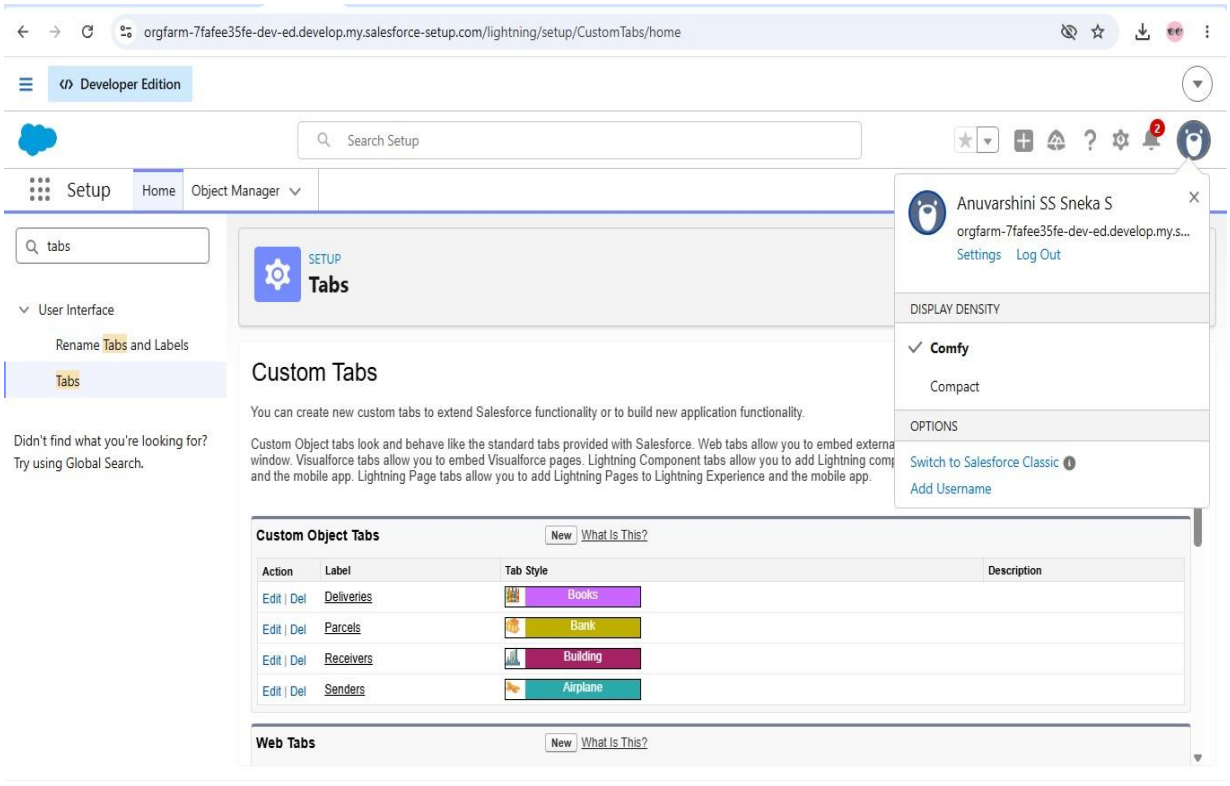
Parcel Object:

The Parcel object is used to maintain information about parcels being booked and tracked. It contains details such as Parcel ID, Status, Weight, Estimated Delivery Date, Sender, and Receiver. The Parcel object is linked with the Sender, Receiver, and Delivery objects so that each parcel can be associated with the corresponding shipment and delivery information



PARCEL OBJECT

TABS



CUSTOM TABS

Project Tracker, Velocity & Burndown Chart:

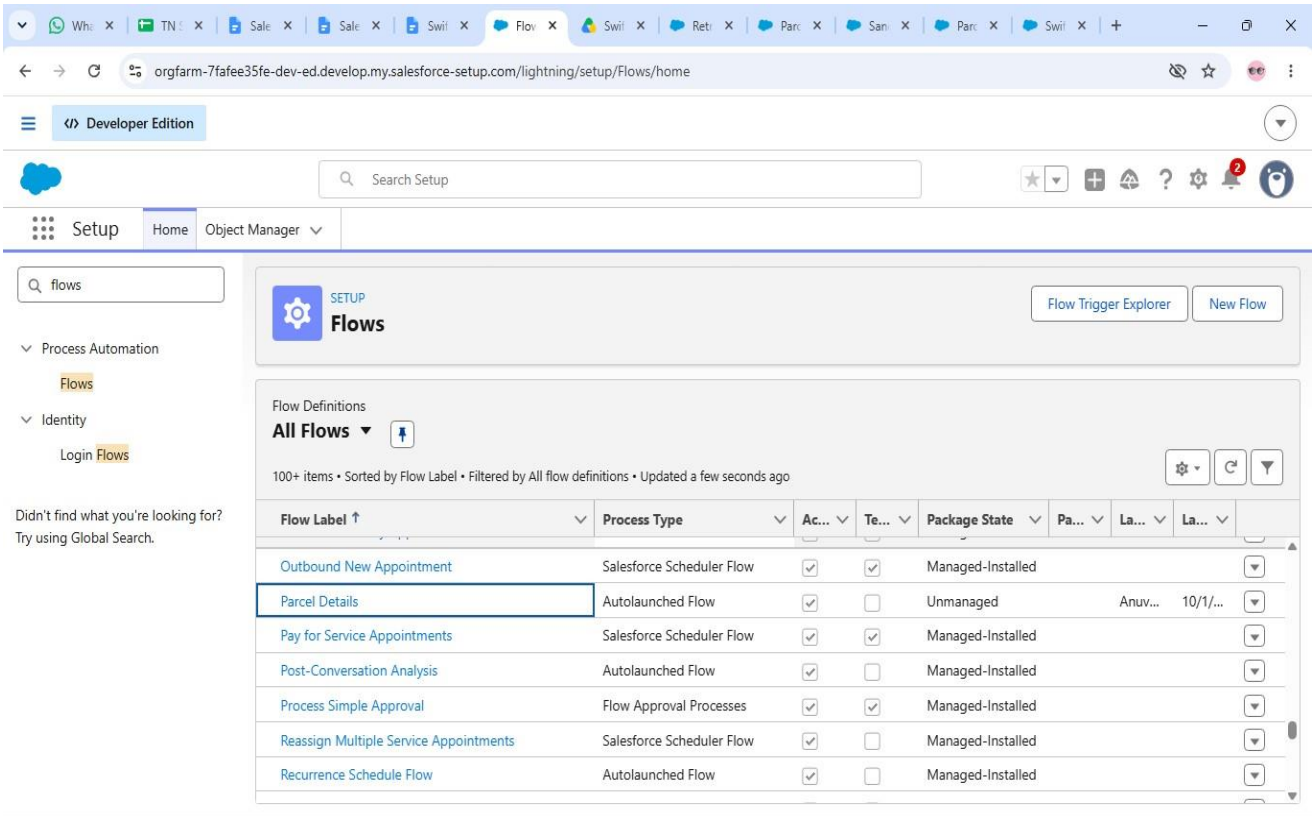
Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members

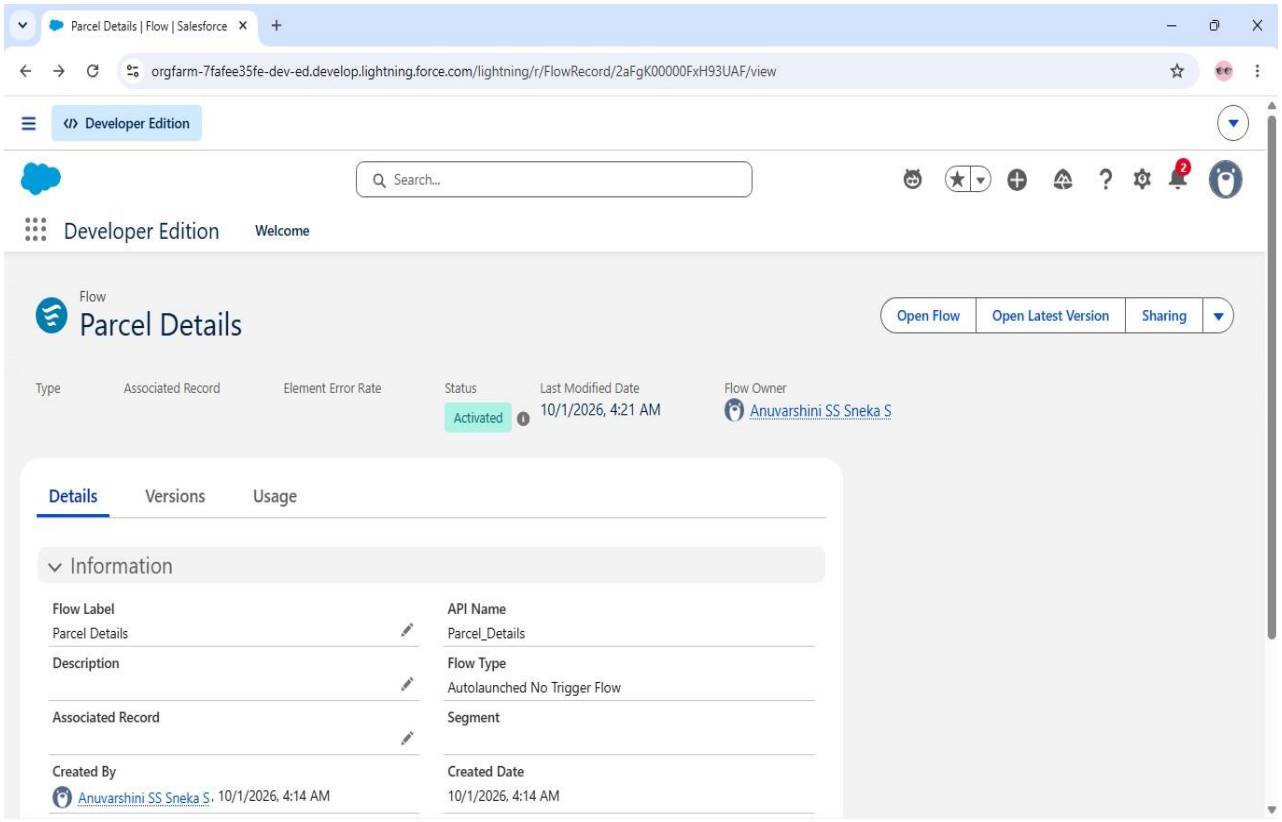
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure the Salesforce environment so that I can build the SwiftShip Tracker parcel management system.	3	High	Member1
Sprint-1	Data Modeling	USN-2	As an admin, I want to create Parcel, Delivery, Sender, and Receiver objects with proper relationships so that parcel and delivery data is stored and managed systematically.	5	High	Member1
Sprint-2	Data Modeling	USN-3	As a staff member, I want dedicated tabs and a Lightning App for SwiftShip modules so that I can easily navigate and manage parcel and delivery operations..	5	High	Member1
Sprint-2	Automation	USN-4	As a system, I want to validate parcel details, mandatory fields, and booking inputs so that invalid or incomplete parcel records are prevented..	3	High	Member2
Sprint-3	Automation	USN-5	As a delivery coordinator, I want automated Flows to update parcel status and delivery information so that parcel processing requires minimal manual effort..	3	High	Member1
Sprint-3	Apex	USN-6	As a system, I want Apex triggers and classes to automatically perform critical parcel and delivery actions so that important processes are handled accurately..	5	High	Member1

Sprint-4	Apex	USN-7	As an inventory/operations manager, I want Batch Apex jobs to process and update parcel or delivery records automatically so that operational data remains accurate and up to date.	4	High	Member1
Sprint-4	Dashboards & AI	USN-9	As a manager, I want dashboards showing parcel status, delivery performance, and key metrics, while Agentforce provides AI-powered tracking assistance, so that I can monitor operations efficiently..	4	Medium	Member2

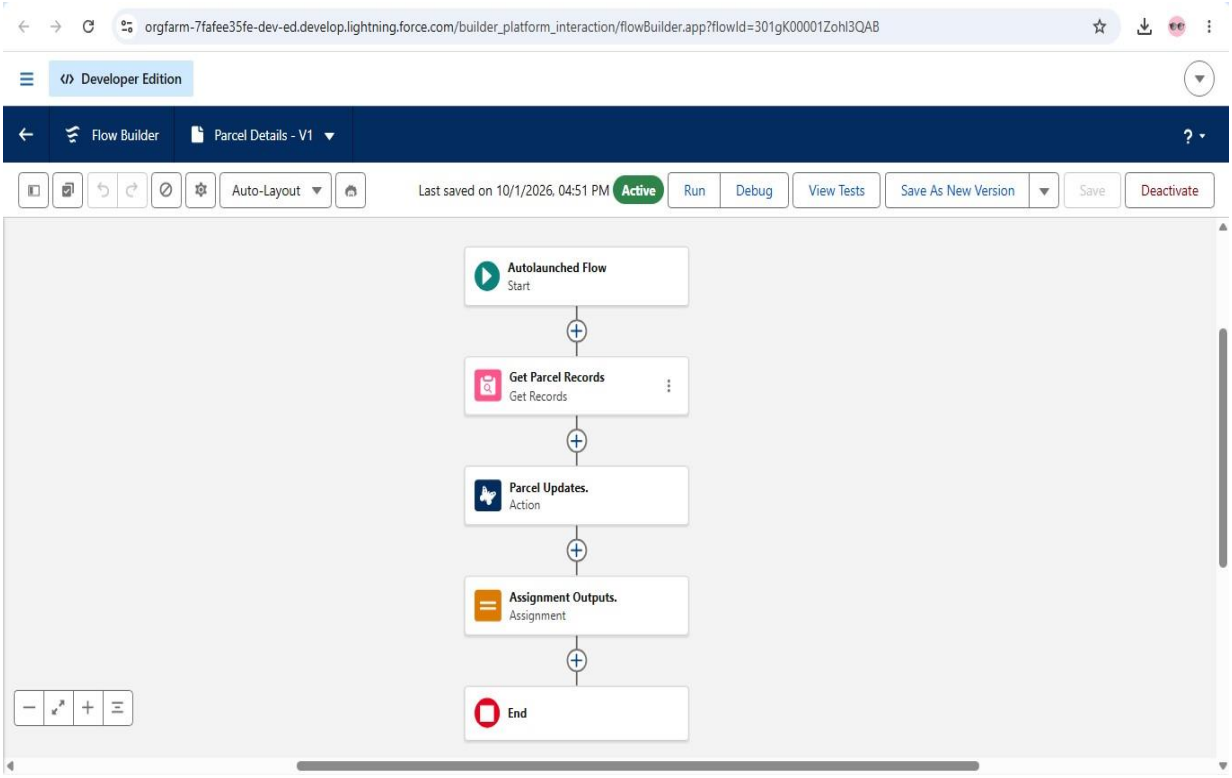
5. Flows

Flows are used to automate parcel-related processes in the Swift Ship Tracker system. They help automate tasks such as parcel processing, status updates, and delivery-related actions without requiring manual intervention.

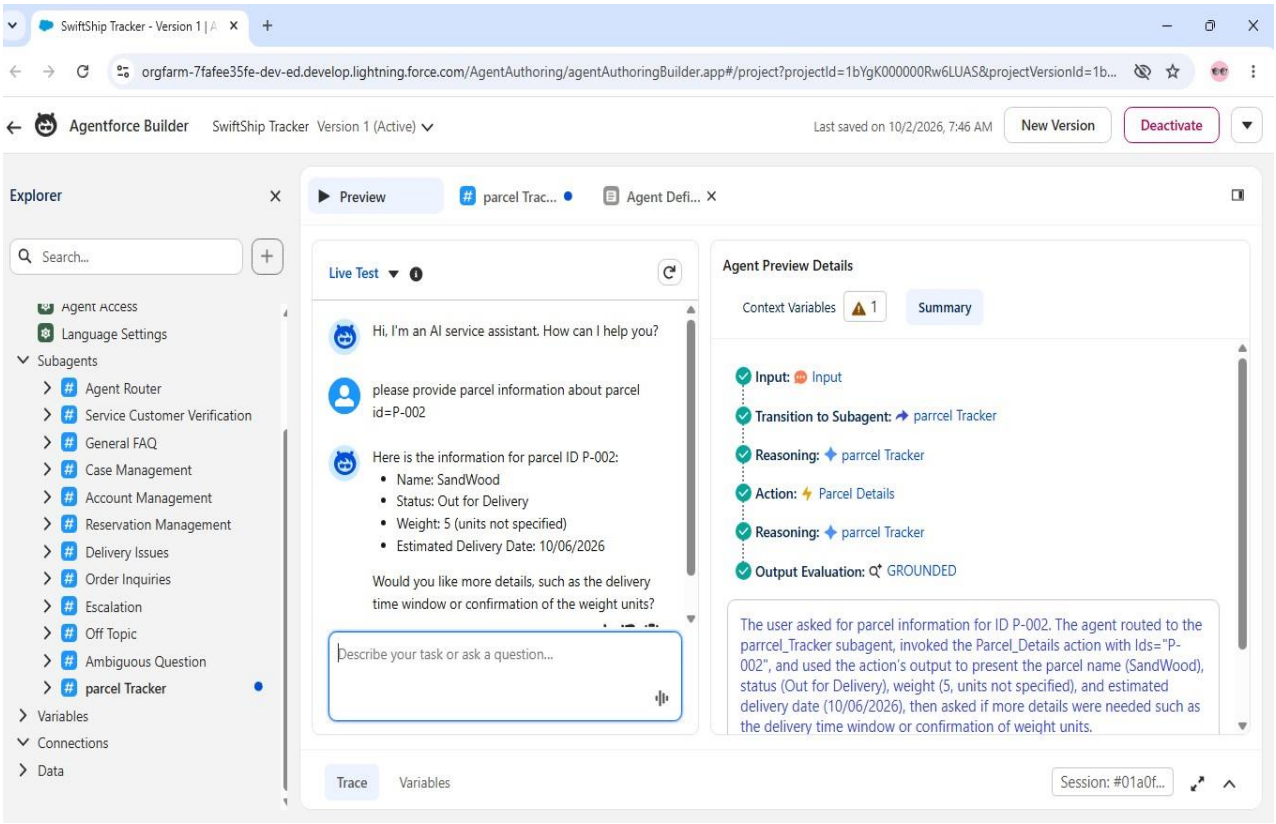

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Developer Edition
Search Setup
Setup
Home
Object Manager
flows
Process Automation
Flows
Identity
Login Flows
Didn't find what you're looking for?
Try using Global Search.
SETUP
Flows
Flow Trigger Explorer
New Flow
Flow Definitions
All Flows
100+ items • Sorted by Flow Label • Filtered by All flow definitions • Updated a few seconds ago
Flow Label
Process Type
Ac...
Te...
Package State
Pa...
La...
La...
Outbound New Appointment
Salesforce Scheduler Flow
Managed-Installed
Parcel Details
Autolaunched Flow
Unmanaged
Anuv...
10/1/...
Pay for Service Appointments
Salesforce Scheduler Flow
Managed-Installed
Post-Conversation Analysis
Autolaunched Flow
Managed-Installed
Process Simple Approval
Flow Approval Processes
Managed-Installed
Reassign Multiple Service Appointments
Salesforce Scheduler Flow
Managed-Installed
Recurrence Schedule Flow
Autolaunched Flow
Managed-Installed



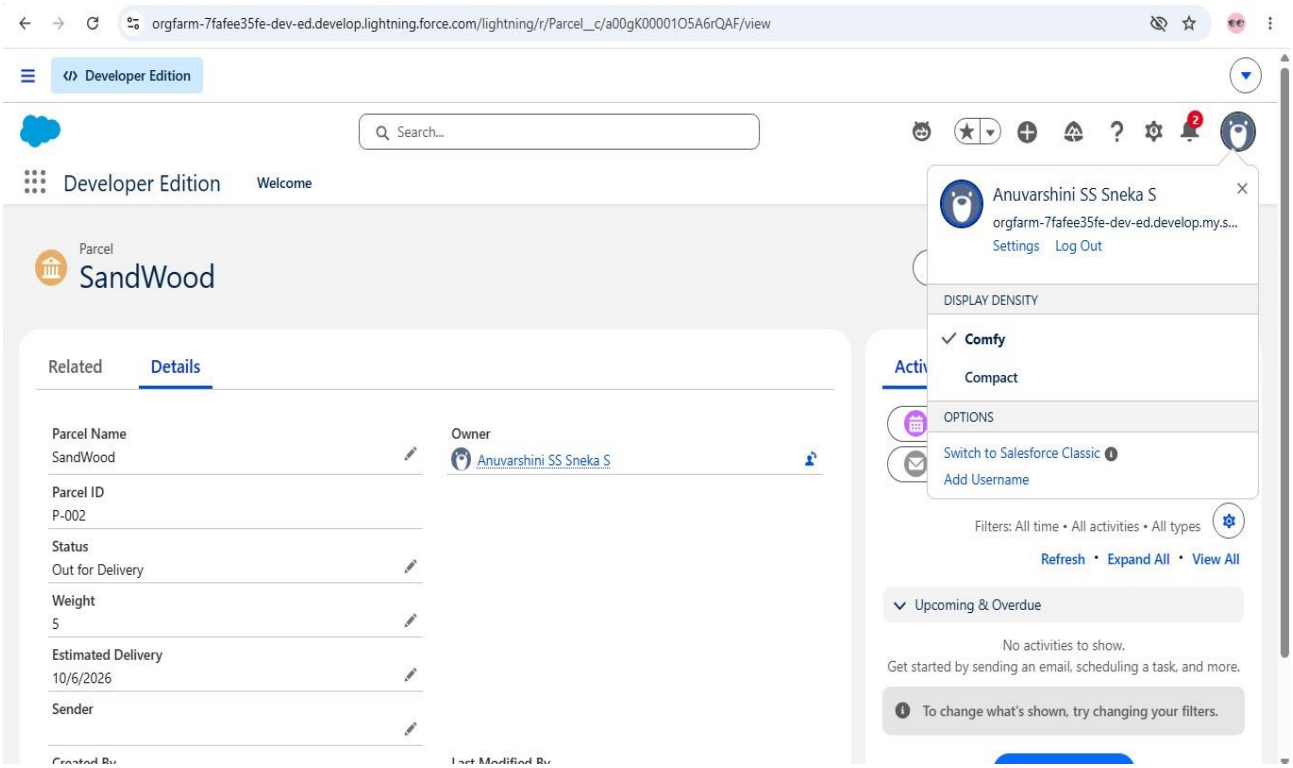
ACTIVATION STATUS



Flow builder for parcel Details



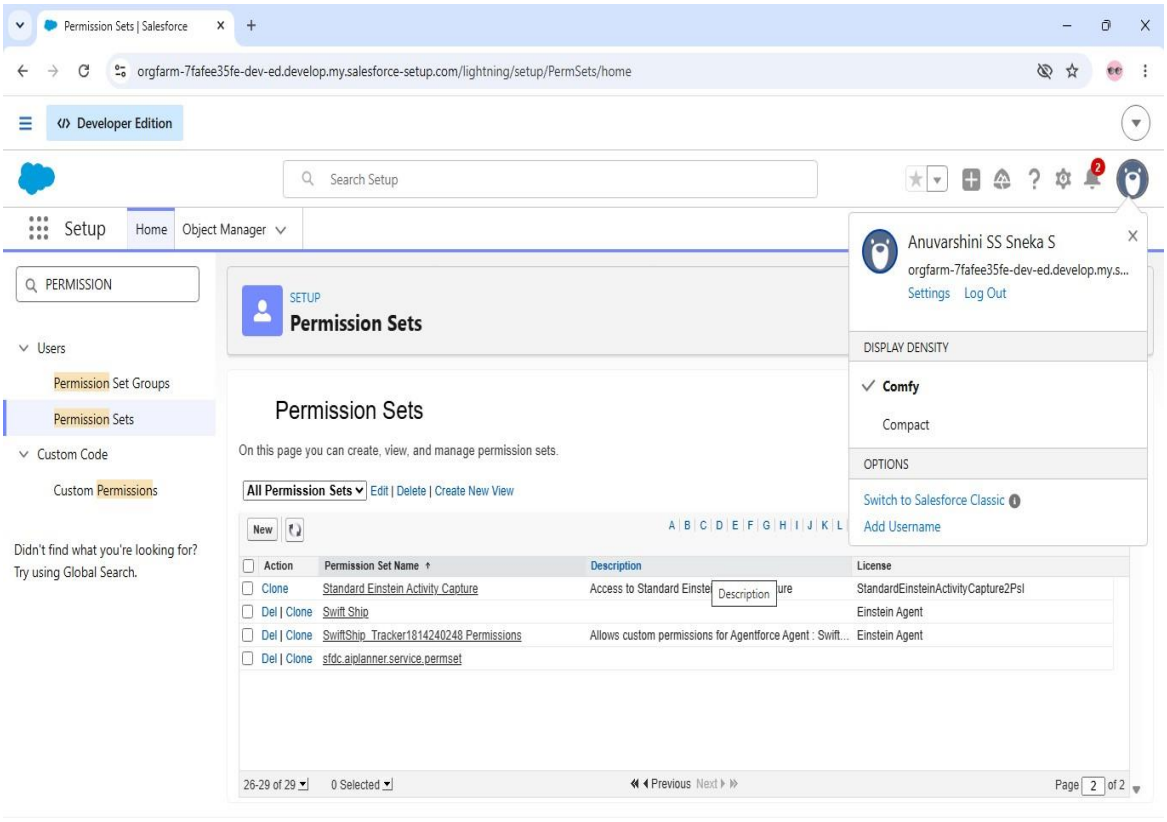
SWIFT SHIP TRACKER OUTPUT

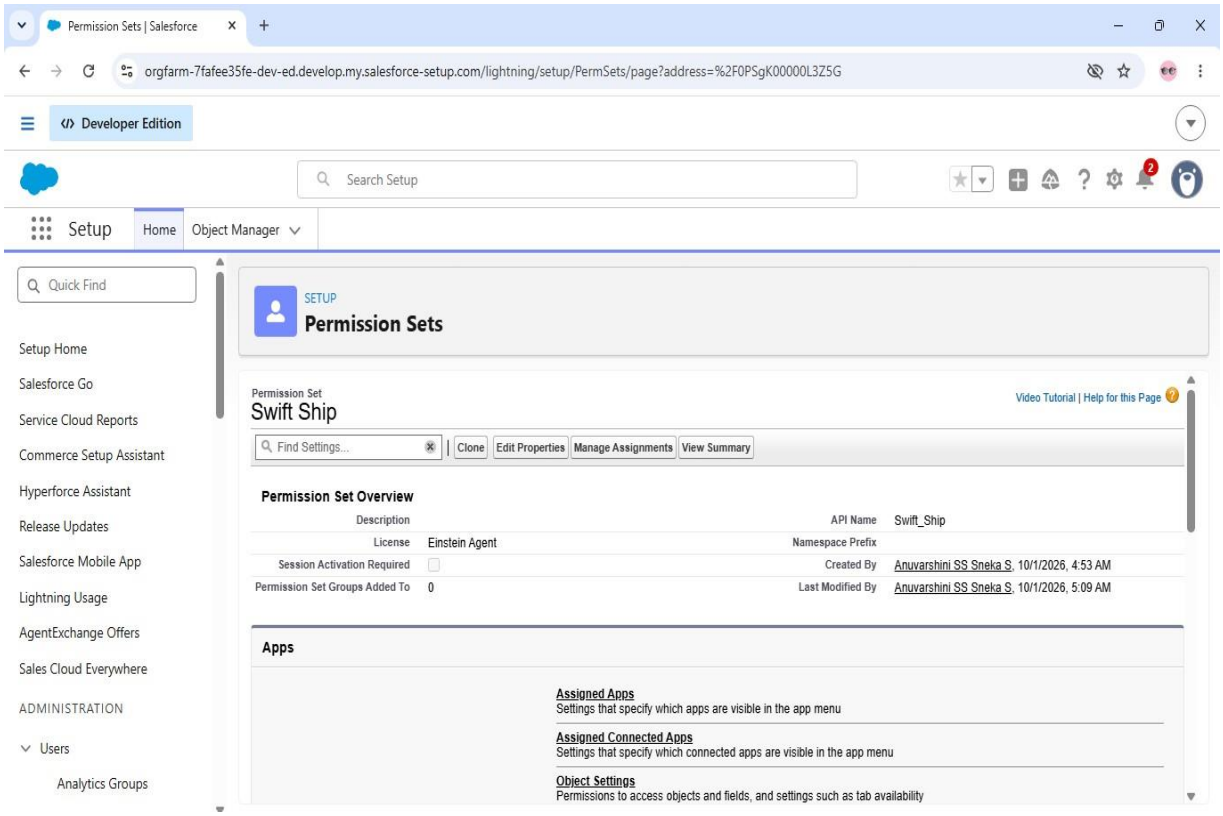

orgfarm-7fafee35fe-dev-ed.develop.lightning.force.com/lightning/r/Parcel__c/a00gK00001O5A6rQAF/view
Developer Edition
Search...
Developer Edition
Welcome
Parcel
SandWood
Related
Details
Parcel Name
SandWood
Owner
Anuvarshini SS Sneka S
Parcel ID
P-002
Status
Out for Delivery
Weight
5
Estimated Delivery
10/6/2026
Sender
Anuvarshini SS Sneka S
orgfarm-7fafee35fe-dev-ed.develop.my.s...
Settings
Log Out
DISPLAY DENSITY
Comfy
Compact
OPTIONS
Switch to Salesforce Classic
Add Username
Filters: All time • All activities • All types
Refresh • Expand All • View All
Upcoming & Overdue
No activities to show.
Get started by sending an email, scheduling a task, and more.
To change what's shown, try changing your filters.

PARCEL NAME

6. PHASE 3 – UI/UX DEVELOPMENT & CUSTOMIZATION

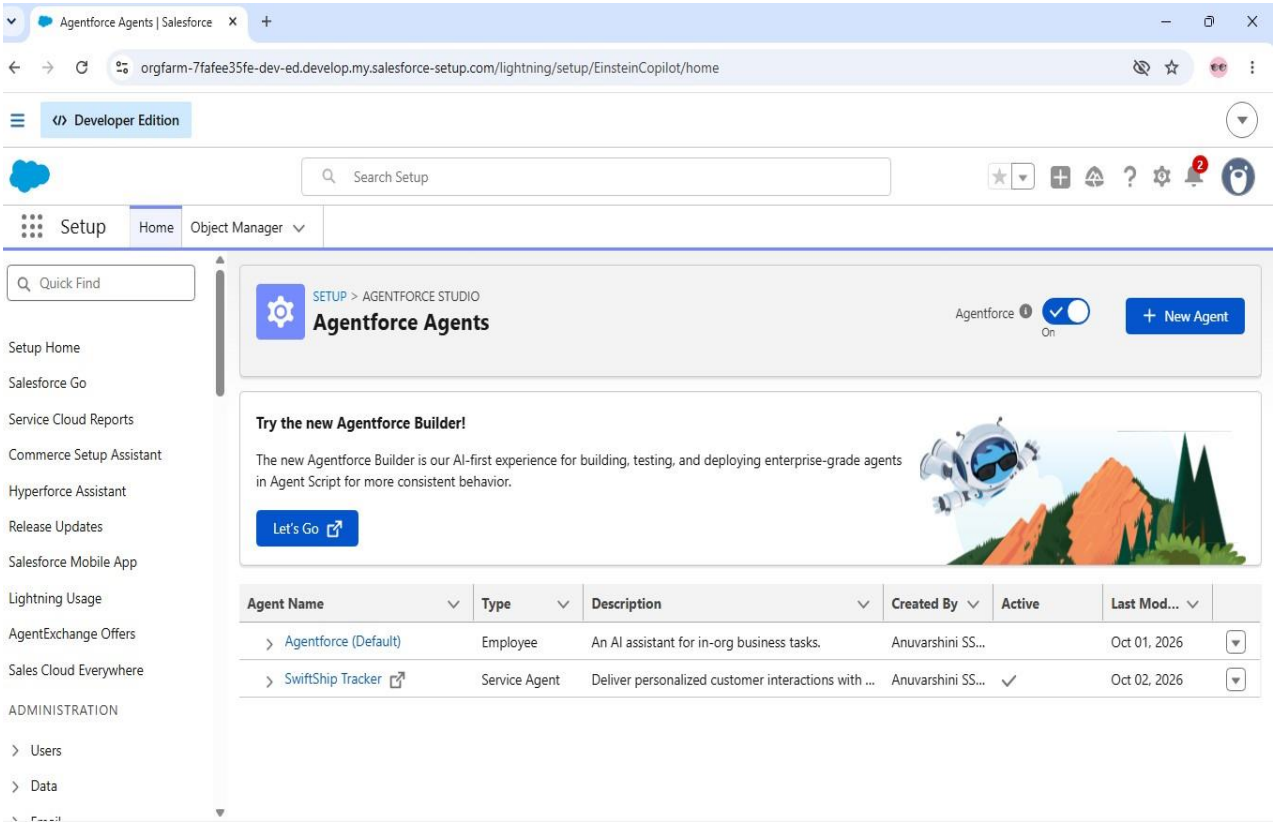
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PERMISSION SETS

7. Phase 4 – Data Migration, Testing & Security



AGENT VERIFICATION

Functional & Performance Testing

Security Testing:

Security configurations were verified using **Profiles, Role Hierarchy, Permission Sets, FieldLevel Security, and Sharing Rules**. These configurations were reviewed to ensure that parcel, sender,

receiver, and delivery information is accessed only by authorized users according to their assigned responsibilities.

8. PHASE 5 – DEPLOYMENT, Documentation & Maintenance

Deployment Strategy

The **Swift Ship Tracker System** was developed and tested in a **Salesforce Developer Edition** environment. All major components, including custom objects, fields, Flows, Apex functionality, Prompt Builder, and security configurations, were configured and verified within the development environment.

Production deployment was not performed as part of this academic implementation. For a realworld implementation, the configured components can be moved to a production Salesforce environment using appropriate deployment methods such as **Sandboxes, Change Sets, Salesforce DX, or DevOps/CI-CD tools**.

Before production deployment, the system should be validated through proper testing to ensure that parcel tracking, delivery updates, automation, AI functionality, and user access work as expected.

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Maintenance Approach

Regular maintenance is important to keep the **Swift Ship Tracker** system stable and useful as parcel management requirements change.

The maintenance activities would include:

- Monitoring parcel booking and delivery processes for any operational issues.
- Checking shipment records and correcting incomplete or inconsistent information.
- Reviewing parcel status and delivery details to ensure they remain up to date.
- Updating the system interface when users require changes or improvements
- Maintaining tracking information based on changing delivery requirements
- Reviewing user access and ensuring that the appropriate users can access the required information.

Periodically checking the overall system behaviour and resolving issues identified during usage.

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Sprint Planning / Schedule

Sprint	Total Story Points	Duration	Sprint Start	Planned End	Story Points Completed	Actual Release
Sprint-1	20	8 Days	11 Apr 2021	18 Apr 2021	20	19 Apr 2021
Sprint-2	20	6 Days	12 May 2023	17 May 2023	—	—
Sprint-3	20	5 Days	14 Nov 2023	18 Nov 2023	—	—
Sprint-4	20	4 Days	11 Dec 2023	14 Dec 2023	—	—

Project Tracker, Velocity & Burndown Chart

The project progress was planned and tracked using sprint-wise story points. The project tracker records the planned story points, sprint duration, start date, planned end date, completed story points, and actual release date. Velocity and burndown tracking can be used to monitor sprint progress and support future estimation.

9.Advantages & Disadvantages

9.1.Advantages

- Centralized Shipment Management – Swift Ship Tracker brings parcel, sender, receiver, and delivery information together in one organized system..
- **Improved Parcel Visibility – Users can easily follow shipment progress and view the current stage of parcel delivery.**
- **Efficient Delivery Handling** – Delivery-related information can be maintained systematically, helping staff manage shipments more effectively..
- **Reduced Manual Work** – The system minimizes repetitive activities involved in maintaining and updating parcel records..
- **Easy Parcel Identification** – Each shipment can be identified using its unique Parcel ID, making information retrieval easier..
- **Better Information Access – Required parcel and delivery details can be accessed quickly from the system**
- **Improved Coordination** – Connecting sender, receiver, parcel, and delivery information helps maintain a clear flow of shipment details..
- **Customer Convenience** – Customers can obtain relevant parcel information without depending entirely on manual communication..

9.2 Disadvantages

- **Data Accuracy Requirement – The system requires accurate and updated parcel and customer information for proper shipment management**
- **Salesforce Configuration Dependency** – System functionality depends on the proper configuration of Salesforce features and automation..
- **AI Response Dependency** – Agentforce responses depend on suitable prompts and the availability of relevant parcel information..
- **Integration Complexity** – Connecting the system with external logistics services may require additional technical configuration..
- **Automation Maintenance** – Changes in delivery processes may require updates to existing flows and automated operations..

10.Conclusion

The **SwiftShip Tracker** demonstrates how Salesforce can be used to create a centralized solution for parcel management and shipment tracking. The system connects **Parcel, Sender, Receiver, and Delivery** information to support parcel booking, delivery monitoring, status updates, and customer communication.

The project uses **Salesforce Lightning, Flows, Apex, Email Alerts, Experience Cloud, Agentforce AI, Prompt Builder, security controls, Reports, and Dashboards** to support the major requirements of the parcel management process. The combination of Agentforce AI, Prompt Builder, and Salesforce Flow also provides a conversational way for users to access shipment information with less dependence on manual support.

Overall, the **SwiftShip Tracker** provides a structured foundation for managing parcel operations and can be further enhanced with additional tracking, automation, and customer-service capabilities.

11.Future Scope

The **SwiftShip Tracker** can be further enhanced to support expanding parcel operations and provide better tracking, automation, integration, and customer services..

- **Delivery Route Planning –** Introduce route-based planning to help delivery teams organize shipments more effectively**.**
- **Priority Shipment Handling** – Add options to identify and manage urgent or priority parcels separately..
- **Delivery Slot Management –** Provide customers with suitable delivery time-slot options based on operational availability**.**
- **Exception Management** – Add features to identify delayed, misplaced, or unsuccessful deliveries and record appropriate actions.
- **Customer Feedback Module** – Enable customers to submit feedback after delivery and use the information to improve service quality.
- **Shipment History –** Maintain a detailed history of previous parcel activities for easier reference and review**.**
- **Operational Alerts** – Provide alerts for important delivery events such as delays, failed delivery attempts, or pending actions..

Performance Monitoring – Introduce additional measures to monitor delivery-agent activities and overall shipment handling efficiency

12.APPENDIX

A. Custom Objects

- Parcel
- Sender
- Receiver
- Delivery

B. Automation

- Record-Triggered Flow – Parcel Status Update
- Record-Triggered Flow – Delivery Information Update
- Scheduled Flow – Shipment Monitoring
- Validation Rule – Parcel Information Validation

C. Standard Objects

- **Account** – Stores customer details.
- **Contact** – Stores contact information.
- **Task** – Tracks assigned activities.

D. Security

- Profiles – System Administrator, Delivery Agent, Support Staff, Customer
- Roles – Administrator, Delivery Manager, Delivery Agent, Customer
- Permission Set – Parcel Tracking Access
- Field-Level Security – Parcel and Delivery Information
- Sharing Rules – Parcel Access for Delivery Users

E. Apex

- **Apex Classes** – Handles custom processing.
- Parcel Record Data Management

F. UI

- **Lightning App** – Provides the main application interface.
- **Lightning Record Pages** – Displays record information.

G. Testing

- **Flow** – Tests automation processes.
- **AgentForce** – Tests AI responses.

- **Security and Permission Validation** – Checks user access.

H. Analytics

- **Reports** – Displays shipment information.
- **Dashboards** – Shows shipment data visually.

Thank You!!